



Analysis of the instability of two-dimensional modes in gaps

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The study of the interaction of gaps with the boundary layer has been going on for years, but several questions remain open. Trying to understand how oscillation modes are influenced by geometry and other parameters is extremely important for industry since the drag produced by a turbulent boundary layer is much greater than that of a laminar boundary layer. Based on an experimental parametric scan, we sought to obtain more details and some explanations about what was happening. To do this, the theory of linear stability was used, using an in-house code. The scan showed that changes in the depth and length of the gap have different effects on the instability of the modes. For a preliminary study, we decided to initially analyze the two-dimensional modes (Rossiter modes). With this study, it was possible to see how the instability evolves as the cavity gets longer or deeper, as well as to understand how the Reynolds number can affect the instability.

I. Nomenclature

C_n	=	Centrifugal mode n
D	=	Cavity depth
f	=	Frequency
L	=	Cavity length
M	=	Mach number
N	=	T-S spatial amplification ratio
R_n	=	Rossiter mode n
Re_D	=	Reynolds number based on cavity depth
Re_L	=	Reynolds number based on cavity length
Re_{δ^*}	=	Reynolds number based on displacement thickness
T-S	=	Tollmien-Schlichting wave
U_∞	=	Free-stream velocity
X	=	Stream-wise distance normalized by displacement thickness
Y	=	Wall-normal distance normalized by displacement thickness
β	=	Span-wise wave number
δ^*	=	Displacement thickness
λ	=	Span-wise wavelength
σ	=	Amplification rate
ω	=	Frequency

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II. Introduction

The shear forces in the boundary layer are characteristic of a dissipative nature. Specifically in an aircraft, these forces contribute to drag in the form of viscous skin friction. Furthermore, a turbulent boundary layer exerts a significant increase in drag. As a consequence, engineering has an economic interest in investigating transition. To illustrate the impact on economic and environmental aspects, Schneider [1] quantifies that a hypothetical reduction of 1% in the total aerodynamic drag of a large commercial aircraft operating over long distances would result in savings of 400,000 liters of fuel and a decrease of 5,000 kg of harmful gases emitted per year. Also, Marec [2] states that about 50% of the total drag of a typical civil transport aircraft resides in viscous drag.

In two-dimensional boundary layers, the fundamental oscillation mode is called the Tollmien-Schlichting (T-S) wave, named after Tollmien [3] and Schlichting [4]. Regarding transition detection, van Ingen [5] and Smith and Gamberoni [6] independently suggested a semi-empirical method based on the linear stability theory, the so-called e^N method. However, other forms of transition may arise, such as transient growth or bypass transition [7]. Many authors investigate the impact that surface imperfections have on T-S growth. Perraud et al. [8] compared the variation of the factor N (ΔN) computed by T-S growth in a boundary layer with and without a gap. Forte et al. [9] performed an experimental investigation to validate the model proposed by Perraud et al. [10]. The authors provided evidence of bypass transition. Beguet et al. [11] revealed the limits for bypass conditions according to the gap length and depth normalized by the displacement thickness, $L/\delta^* \geq 18$ and $D/\delta^* \geq 2$, respectively. Crouch and Kosorygin [12] performed experimental correlations investigating step, regular protrusion, and gap effects on the boundary layer transition. Crouch et al. [13] performed experimental ΔN correlations for cavity influence on transition with different sizes. In one particular run that underwent bypass transition, they observed a frequency greater than expected for the T-S unstable range for this flow configuration. Zahn and Rist [14] numerically investigated the impact of deep gaps on the boundary layer transition by comparing the growth of T-S waves (ΔN) introduced in the flow for runs with and without the gap. They concluded that a stationary acoustic wave inside the gap had a substantial impact on the growth of T-S waves (ΔN). Mathias and Medeiros [15] compared the spatial evolution of wavepackets over the gap with the smooth flat plate under the same conditions. The wavepackets showed a local change over the gap, but further downstream, such distortion was attenuated, except for a phase shift. The frequency content revealed a change in amplitude for higher frequencies, whereas the effect was negligible in the lower range of frequencies.

Earlier investigations have shown the presence of instabilities related to the flow over a cavity, as reported by Krishnamurty [16], Plumblee et al. [17], and Rossiter [18]. The latter empirically proposed a physical mechanism to explain the acoustic radiation marked by a distinct frequency. The collision of vortices shed from the cavity leading edge into the cavity trailing edge generates acoustic waves that destabilize the mixing layer at the leading edge, completing the feedback cycle. Sarohia [19] carried out experiments to determine the threshold of the onset of cavity oscillation. Yamouni et al. [20] performed global stability analysis and observed an overshoot in the amplification of unstable Rossiter modes when interacting with the standing waves from Plumblee et al. [17]. Mathias and Medeiros [21] studied the dependency of the Mach number and the displacement thickness on the gap stability, concluding that the incoming boundary layer plays a role in selecting which Rossiter mode will occur, whereas the Mach number increases the acoustic energy transfer and the temporal growth rate. Furthermore, the cavity flow presents three-dimensional oscillatory modes associated with centrifugal instability. Brés and Colonius [22] numerically investigated the stability of three-dimensional perturbations in two-dimensional flow over a cavity. Their results indicated the presence of unstable modes with wavelengths about the cavity depth, differing from each other in the frequency range. According to them, this instability arose due to a non-acoustic nature attributed to the centrifugal mechanism. They also found a low dependence of this mode on the Mach number, while the Reynolds number exerted influence on the instability and hierarchy of the dominant modes. De Vicente et al. [23] and Meseguer-Garrido et al. [24] investigated the centrifugal instability in flow regimes in which the Rossiter mode was not present. The authors constructed the neutral stability diagram and amplification curves for different span-wise wave numbers and Reynolds numbers. Sun et al. [25] compared the results from linear stability analysis and three-dimensional DNS. The authors advised that cavities with larger aspect ratios had relevant non-linear interactions between the 2D and 3D modes, diverging from the linear assumptions of stability analysis. Mathias and de Medeiros [26] numerically investigated the interaction between Rossiter and centrifugal modes. The authors noted that in the 3D runs, the peaks in the spectrum associated with the Rossiter were less sharp than in the 2D runs. They attributed this to a span-wise modulation of the Rossiter due to the centrifugal modes. According to them, such motion could potentially alter the base flow and consequently its stability. Victorino et al. [27] computationally found a connection between this bypass transition and unstable modes in the cavity, such as Rossiter and centrifugal modes. In this study, the DNS exhibited a non-linear interaction between the centrifugal and Rossiter modes leading to transition. For this reason, in this work, we will explore how different parameters can impact two-dimensional instability.

III. Methodology

A. Experimental Methodology

The experiments were performed at LANT (Low Acoustic Noise and Turbulence), a closed-circuit wind tunnel at EESC-USP. It has a square test section with dimensions $1 \text{ m} \times 1 \text{ m}$ and a length of 3 m . More details about the wind tunnel are available at Amaral et al. [28]. The wind tunnel features low acoustic noise and turbulence level in the free stream, with a value of $Tu = 0.036\%$ (bandwidth 1-1024 Hz) measured with the test section empty, and free-stream of 20 m/s . The experiment was performed in a flat plate which has an insert that houses a system that changes the geometry of the surface imperfection, i.e. it is possible to form a smooth flat plate, a bump, or a gap. This system is positioned 600 mm from the leading edge of the model. The gap is 30 mm long in the streamwise direction and has a span of 600 mm .

The velocity fluctuations were measured using Hot Wire Anemometry (HWA). The anemometer used is the AN-1002 model from the A.A.Lab system, operating in CTA (Constant Temperature Anemometry) mode. The probe used as a sensitive element is the Dantec Dynamic model 55P15, indicated for boundary layer measurements. The probe can be moved to the desired point using a high-precision 3D Cartesian traverse, with a movement resolution in the normal direction to the wall (y) of $6.25 \text{ } \mu\text{m}$. The former, DAq-USB-6002 from National Instruments, is used for average velocity measurements and features 16-bit resolution and a sampling rate of up to 50,000 samples per second. The anemometer voltage values are converted into velocity through calibration with velocity values measured by a static Pitot positioned inside the test section. The latter consists of a PXI-4496 board with 24-bit resolution installed in a PXI-1042Q chassis, both from National Instruments and, due to its higher resolution, it is responsible for acquiring the voltage fluctuation signal.

B. Numerical Methodology

We use Direct Numerical Simulation (DNS) and a time-stepping algorithm based on the Arnoldi method [29] to approximate the eigenvalues and eigenfunctions for the modal analysis. Both codes were developed in-house and are open-source*. The DNS approximates a solution for the compressible Navier-Stokes Equations (NSE), given by Eqs. 1, 2, 3, 4, and 5.

$$\frac{\partial \rho}{\partial t} = -\rho \frac{\partial u_i}{\partial x_i} - u_i \frac{\partial \rho}{\partial x_i} \quad (1)$$

$$\frac{\partial u_j}{\partial t} = -u_i \frac{\partial u_j}{\partial x_i} - \frac{1}{\rho} \frac{\partial p}{\partial x_j} + \frac{1}{\rho} \frac{\partial \tau_{ij}}{\partial x_i} \quad (2)$$

$$\frac{\partial e}{\partial t} = u_i \frac{\partial e}{\partial x_i} - \frac{p}{\rho} \frac{\partial u_i}{\partial x_i} + \frac{\tau_{ij}}{\rho} \frac{\partial u_j}{\partial x_i} - \frac{1}{\rho} \frac{\partial q_i}{\partial x_i} \quad (3)$$

Where the viscous stress tensor and heat flux are given by:

$$\tau_{ij} = \frac{\mu}{Re} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \frac{\partial u_k}{\partial x_k} \right) \quad (4)$$

$$q_i = -\frac{\mu}{(\gamma - 1) Re Pr Ma^2} \frac{\partial T}{\partial x_i} \quad (5)$$

The variables represented by scalars such as density, ρ , energy, e , temperature, T , and pressure, p . The velocity vector is represented as $\vec{V} = u\hat{i} + v\hat{j} + w\hat{k}$, and all variables are non-dimensionalized by their reference scales. The non-dimensional parameters Re , Pr , and M represent the Reynolds, Prandtl, and Mach numbers, respectively. The domain is represented by a Cartesian, structured, and stretched mesh. Buffer zones are used to dissipate reflexive oscillatory motion near open boundaries by increasing node spacing and reducing the accuracy order of the spatial derivatives approximation. The spatial terms of the equation are approximated using a spectral-like differentiation scheme based on [30] and [31]. A spatial anti-aliasing filter adapted from [32] is employed to attenuate motion not resolved by the mesh. Temporal integration is performed using a fourth-order Runge-Kutta method. The upstream boundary condition assumes a uniform flow with constant energy and a zero streamwise pressure gradient. The outlet and outflow boundaries are subjected to zero-second derivatives and a prescribed pressure. A free-slip wall model

*https://github.com/marlonmathias/GATT_DNS

with zero wall-normal streamwise velocity derivative ($\partial U / \partial y$) is employed to represent the incident flow at the bottom boundary. The flat plate energy is kept at a constant isothermal condition, while no-slip and no-penetration boundary conditions are imposed on the velocity field. A zero pressure gradient is also enforced in the normal direction. In the globally unstable flows, the selective frequency damping can be turned on inside the domain, implemented according to [33]. Such an artifice is necessary to obtain a time-invariant solution or base flow. The domain decomposition for parallel execution is performed as described in [34].

As the flow over the gap has significant non-parallel effects, the linear stability analysis is performed using the global stability method. The eigenvalues and associated eigenfunctions of the Jacobian matrix are approximated using a time-stepping method based on [29, 35]. In summary, this method adds a small perturbation to a previously obtained base flow and recursively runs the DNS solver for a defined period. After each iteration, a new orthogonal disturbance is generated for the next step, adding a new entry to a Hessemberg matrix H . After a sufficient number of iterations, the complex eigenvalues and eigenvectors of H are good representatives for those of the original problem. The Real part of the eigenvalues corresponds to the temporal growth, where positive values indicate instability. The imaginary part of the eigenvalues is related to the circular frequency of the mode. The eigenfunction is a projection of a particular phase of the spatial shape of the mode. For details about the implementation, accuracy, validation, and method input parameters, further reading is available in [36].

IV. Experimental Results

The motivation for the present numerical study comes from an experimental observation in our wind tunnel corroborating with previous reports. The experiment consisted of measuring the fluctuation of velocities with the hot wire probe positioned downstream of the trailing edge of the cavity at a physical distance from the wall corresponding to the displacement thickness of the boundary layer on the leading edge of the cavity $y = \delta_g^*$ for the smooth case, that is, without cavity. This distance was chosen given that the Rossiter mode eigenfunction presents a peak in this region.

A list of cavity depths was generated from the results by Crouch et al. [37]. The experiment was conducted with constant speed while the depth of the cavity changed. Figure 1 displays the RMS contours of the collected time series for different cavity depths and velocities. Contours are presented in a logarithmic scale, as values span many orders of magnitude.

Table 1 Parameters used in the experiment to map the oscillation regimes caused by the cavity.

U_∞ [m/s]	δ_g^* [mm]	L/δ_g^*	$Re_{\delta_g^*}$
7.1	1.92	15.6	809
9.0	1.71	17.6	910
9.5	1.66	18.1	935
10.0	1.62	18.6	960
10.6	1.58	19.0	986
11.2	1.54	19.5	1011
13.5	1.40	21.5	1111
17.5	1.23	24.4	1261
22.1	1.10	27.2	1409
28.2	0.98	30.6	1585

Fig. 1 also presents data from the Literature. The abrupt transition boundaries (*bypass*) stipulated by Crouch et al. [37], are shown by the blue dashed line. Additionally, the impact on the transition site measured by Crouch et al. [37] is displayed with the circular markers. The variable ξ represents the normalized transition length, a parameter defined by Crouch et al. [37] to quantify the displacement of the transition site caused by the cavity. In short, a small impact by the cavity on the transition is indicated by a value $\xi \rightarrow 1$, while a large impact is characterized by $\xi \rightarrow 0$. According to Crouch et al. [37], circular markers filled in red indicate an abrupt transition (*bypass*). As some parametric regions presented speed fluctuations marked by intermittency, we calculated the intermittency factor given by the variable γ

according to the methodology used by Veerasamy and Atkin [38].

Fig. 1 indicates several important information regarding the cavity problem. Firstly, we verified that the formation of the Rossiter phenomenon is marked by intermittency. The black contour line indicates $\gamma = 0.05$, it was defined as the boundary for the development of the phenomenon. Below and to the left of the lower border, with the ends at $(L/\delta^*; D/\delta^*) = (15, 6; 4, 1)$ and $(30, 2; 1, 0)$, a progressive increase in the RMS of the fluctuation signal is noted the closer to the border. From this boundary of 5% of intermittency, a jump of an order of magnitude in the RMS of the signal. In turn, the magenta contour line presents the 95% border of intermittency, that is, the parametric region where the Rossiter phenomenon is developed and dominates the flow. This boundary is related to the abrupt transition region (*bypass*) measured by Crouch et al. [37]. One of the differences observed is the presence of an upper boundary of the formation of the Rossiter phenomenon, marked by the 5% line with ends at $(L/\delta^*; D/\delta^*) = (15.6; 6, 4)$ and $(20, 1; 13, 3)$. For depths greater than this limit, another mode of oscillation, notably weaker than the Rossiter, dominates, however, with greater amplitude than the laminar regime.

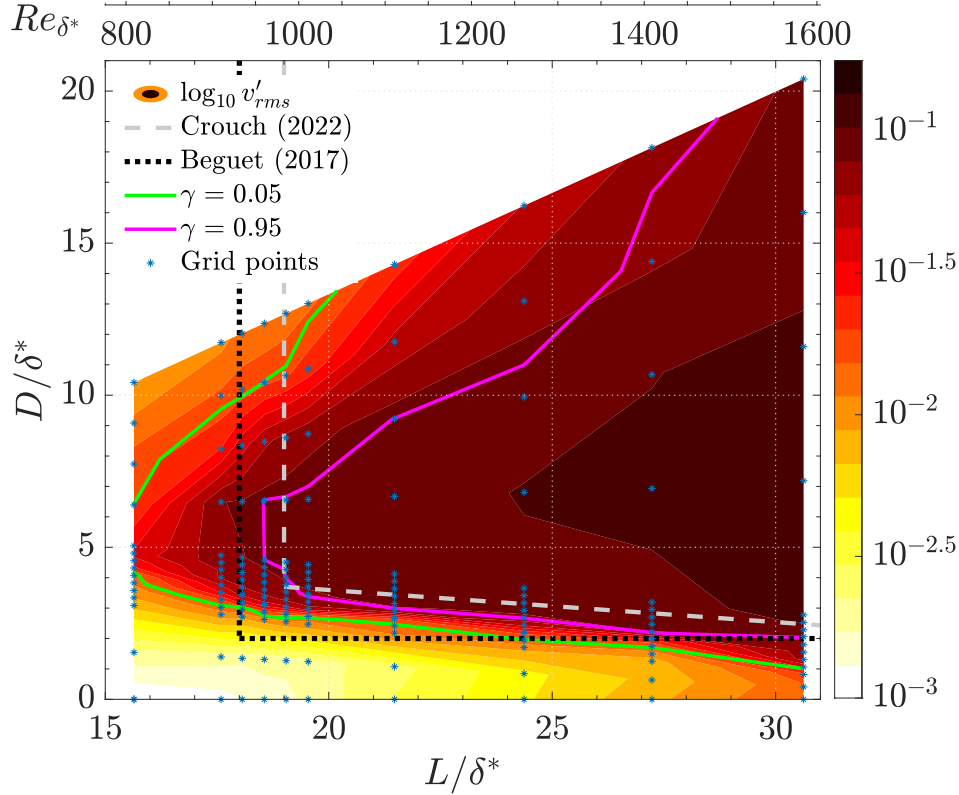


Fig. 1 Contours of the velocity fluctuation RMS in logarithmic scale right downstream of the cavity.

The contours suggest that this behavior occurs for longer cavities, however, up to the investigated parameter space, the boundary appears to behave proportional to the cavity length.

V. Numerical Results

Following the experimental results shown in Fig. 1, we used our code to carry out an LST analysis for all cases at $Re_{\delta^*} = 600$, and Mach number equal $Ma = 0.5$. The wavenumber is from $\beta = 0$ to $\beta = 1$. Using this β range is possible to identify both Rossiter and centrifugal modes. In Fig. 2, the map for different gap geometries is present, this map is the first step for this study, present in [36]

In Fig. 2, it's possible to see three different and progressive regimes. These regimes start from fully stable cases to two- and three-dimensional unstable cases. The first is the regime denoted by green circles. For these cases, all modes are stable. The second regime is denoted by blue squares when only three-dimensional modes are unstable. Besides the marker, we have written the type of unstable modes. In this case, the only three-dimensional unstable modes are the centrifugal modes. The last region, denoted by red triangles, is dominated by two and three-dimensional unstable modes.

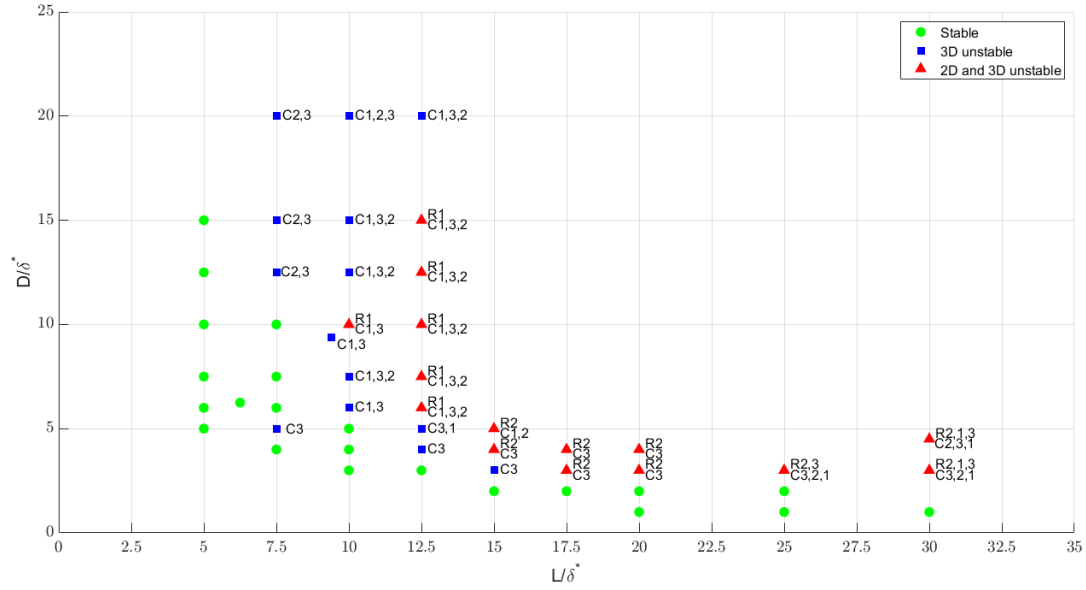


Fig. 2 Map of modes observed by LST

Centrifugal mode numbering follows the same scheme of Brés and Colonius [22], while Rossiter mode numbering denotes the number of vortices present at once along the mixing layer.

In Fig. 2, it's possible to see a progression in the instability. For a line of constant D/δ^* , it's possible to see the instability changing. For shorter gaps, we have all modes stable. The three-dimensional modes become unstable, followed by the two-dimensional modes. For a longer gap, the figure shows a lot of unstable modes. For a line at constant L/δ^* , we can see a similar effect at lower L/δ^* , in long cavities, both modes become unstable simultaneously.

Now, analyzing only the two-dimensional modes (Rossiter), in Fig. 3, we have a contour map for each Rossiter mode. We can note that in the case of Rossiter 1 (Fig. 3a), which is present in most gaps, this mode has a behavior similar to that observed in the experiment. We can notice that as we advance in the map to longer cavities, we begin to have greater amplification in shallower gaps. Meanwhile, as we move into deep gaps, they appear to go beyond an upper limit of amplification rate and decay as the gap becomes deep. Rossiter 2 (Fig. 3b), in consonance with the literature, was observed only for longer gaps, presenting higher amplification rates than the Rossiter 1 modes, dominating the larger cavities of this regime. Finally, Rossiter 3 (Fig. 3c) was only observed for the longest gaps in our parametric range. Furthermore, it is possible to observe that the frequency appears to vary as we change the geometry of the cavity.

To exemplify the different types of gaps in the map, we will analyze the cases along the $D/\delta^* = 3$ line. For the amplification figures, zero represents stable modes. Fig. 4 shows separate frequency and amplification lines. In the figure on the left (Fig. 4a), we can observe the behavior of Rossiter modes 1, 2, and 3. Notice that the amplification of Rossiter mode 2 is more influenced by the gap length. However, the frequency of Rossiter mode 1 tends to increase for long gaps ($L/D > 6$). Rossiter 3 becomes unstable only for the larger gaps with $L/D > 9$. In turn, for the analyzed parametric space, as the cavity becomes longer, the frequency of all Rossiter modes tends to decrease until a minimal, close the $L/D = 8$ (Fig. 4b).

Now, for the depth sweep analysis for a length of $L/\delta^* = 15$, in Fig. 5b. We can notice that unlike what was found for the length sweep, varying the depth has a different impact on the frequency of Rossiter. The frequency passes through a maximum and then decreases as depth increases. In Fig. 5a, we can see a very different behavior for Rossiter 1 and 2 compared to the result presented in 4a. As the depth increases, the instability increases until it reaches a maximum and decreases in sequence, becoming stable for the deeper gaps, modes 1 and 2 reach their peaks at different depths ($7.5D$ and $12.5D$, respectively).

The last parameter is the variation of Re_{δ^*} . For this, we chose two types of gaps, one long and shallow and one short and deep Fig. 6. Initially, analyzing the short and deep cavity, we can notice the presence of only the first Rossiter mode, as Re_{δ^*} increases, as shown in Fig. 6b, the frequency initially increases, until reaching a plateau. The same

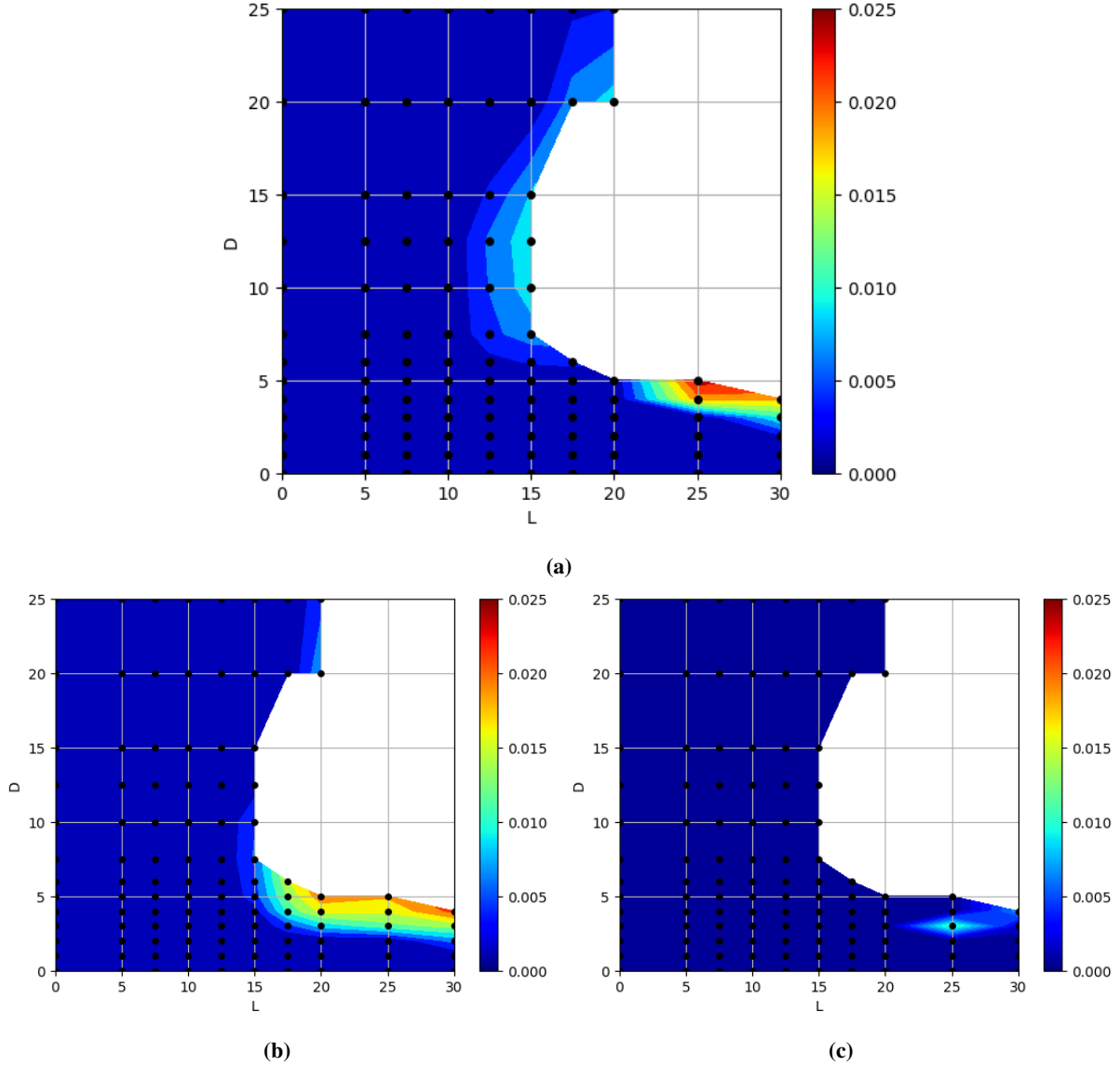


Fig. 3 Map of amplification and frequency of Rossiter mode 1 for different gaps. The color indicates the instability rate (σ) and the diameter indicates the frequency (ω). Figure (a) Rossiter 1, (b) Rossiter 2, (c) Rossiter 3. Zero represents stable mode.

occurs for Rossiter modes 1 and 2 in the larger gap. Moving on to the amplification analysis, in Fig. 6a, we can see that for both cavities, the initial most unstable mode tends to have an initial increase in the amplification, but only the larger gap presents a second Rossiter mode for bigger Re_{δ^*} . An interesting result is the amplification of the Rossiter 2 in the long gap. With the continuous increase of the Re_{δ^*} , the Rossiter 2 becomes more unstable than the Rossiter 1.

VI. Conclusion

This numerical study was motivated by experimental results obtained in the group's low-turbulence wind tunnel, followed by a literature analysis. Our numerical analysis has allowed us to sweep a broader parametric range when compared to our experiments.

Based on our parameter space, we have created an instability map containing unstable two- and three-dimensional modes. For an initial analysis, only two-dimensional modes were chosen, since they present a threshold similar

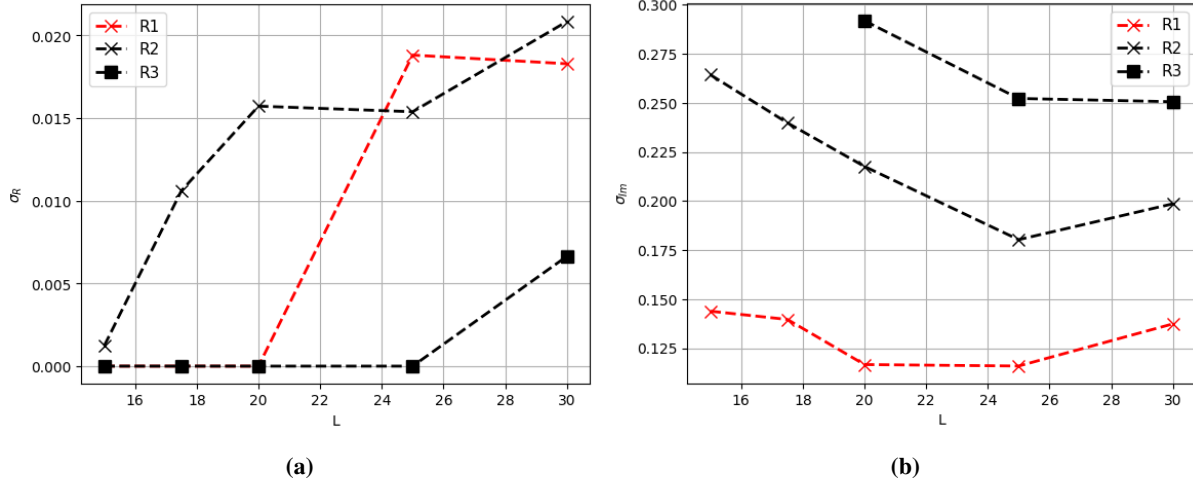


Fig. 4 Scan of the parameter space along the line of $D/\delta^* = 3$. (a) Amplification rate for different gap lengths, Zero represents stable or nonexistent mode and (b) frequency for different gaps

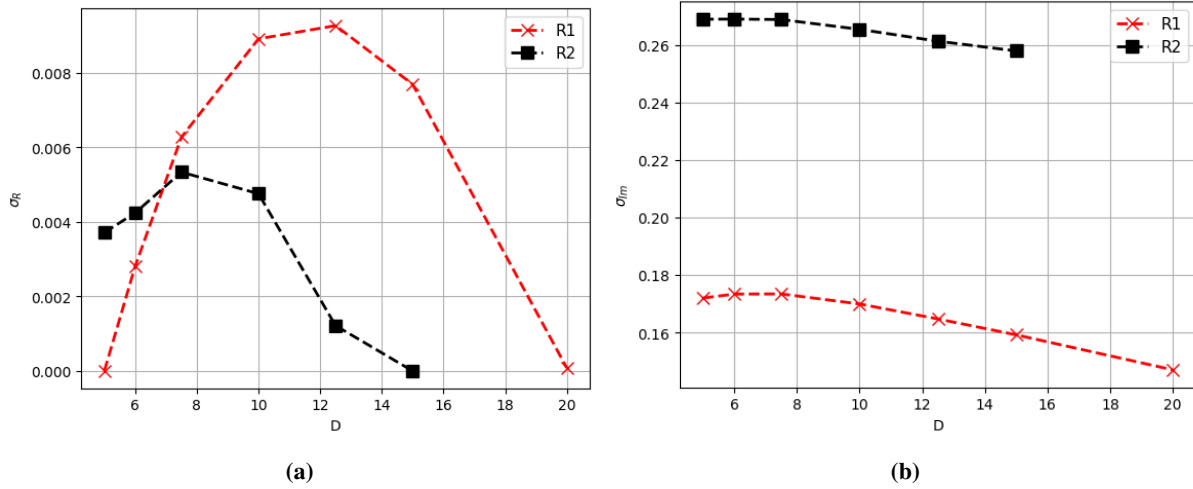


Fig. 5 Scan of the parameter space along the line of $L/\delta^* = 15$. (a) Amplification rate for different gap lengths, Zero represents stable or nonexistent mode and (b) frequency for different gaps

to literature results for boundary layer transition, as well as previous results from the group, that indicate that two-dimensional modes tend to have a greater impact on the transition of the boundary layer.

Analyzing the Rossiter modes found in the gaps, it was possible to note that different types of cavities present different instability regimes. As the depth increases, the instability grows until it reaches a maximum and then decreases again. For the regime where the length increases, the instability of the Rossiter 2 modes tends to increase, with the Rossiter 3 mode appearing for longer gaps. From the sweep of Re_{δ^*} , we noticed a greater influence on unstable modes; however, they tend to reach a plateau as Re_{δ^*} increases.

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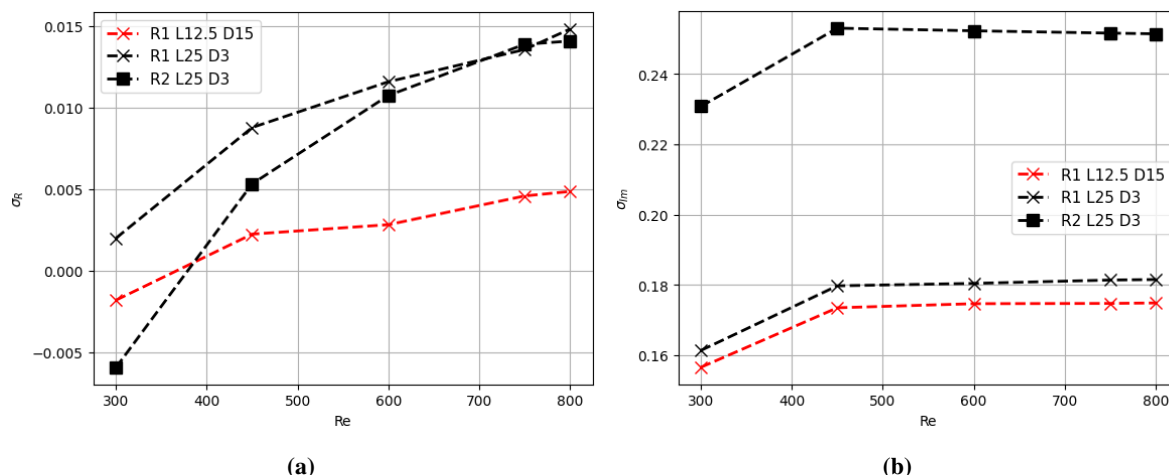


Fig. 6 Scanning the parameter space for two different gaps. In the figure on the left (a) we have the amplification rate for different Reynolds numbers and in figure (b) we have the frequency for different Reynolds numbers.

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References

- [1] Schneider, W., "The Importance of Aerodynamics in the Development of Commercially Successful Transport Aircraft," *Aerodynamic Drag reduction technologies: proceedings of the CEAS, DragNet European Drag Reduction Conference 19 - 21 June 2000, Potsdam, Germany*, 2001, pp. 9–16.
- [2] Marec, J.-P., "Drag reduction: a major task for research," *Aerodynamic Drag reduction technologies: proceedings of the CEAS, DragNet European Drag Reduction Conference 19 - 21 June 2000, Potsdam, Germany*, 2001, pp. 17 – 28.
- [3] Tollmien, W., "The production of turbulence," *National Advisory Committee for Aeronautics - Technical Memorandum no. 609*, 1931.
- [4] Schlichting, H., "Laminare Strahlausbreitung," *ZAMM - Journal of Applied Mathematics and Mechanics / Zeitschrift für Angewandte Mathematik und Mechanik*, Vol. 13, No. 4, 1933, pp. 260–263. <https://doi.org/10.1002/zamm.19330130403>.
- [5] van Ingen, J. L., "A suggested semi-empirical method for calculation of the boundary layer transition region," *Technische Hogeschool Vliegtuigbouwkunde - Report V.T.H.-74*, 1956.
- [6] Smith, A. M. O., and Gamberoni, N., "Transition, Pressure Gradient, and Stability Theory," Tech. Rep. ES 26388, Douglas Aircraft Division, 1956. URL <https://engineering.purdue.edu/~aae519/BAM6QT-Mach-6-tunnel/otherpapers/smith-amoeN-douglas-es26388-1956.pdf>.
- [7] Saric, W. S., Reed, H. L., and Kerschen, E. J., "BOUNDARY-LAYER RECEPTIVITY TO FREESTREAM DISTURBANCES," *Annual Review of Fluid Mechanics*, Vol. 34, No. 1, 2002, pp. 291–319. <https://doi.org/10.1146/annurev.fluid.34.082701.161921>, URL <https://doi.org/10.1146/annurev.fluid.34.082701.161921>.
- [8] Perraud, J., Arnal, D., Séraudie, A., and Tran, D., "Laminar-Turbulent Transition on Aerodynamic Surfaces with Imperfections," *RTO-AVT-111 Symposium*, , No. October 2004, 2004, pp. 4–7. <https://doi.org/10.13140/RG.2.1.3532.1364>.
- [9] Forte, M., Perraud, J., Séraudie, A., Beguet, S., and Casalis, L. G. G., "Experimental and Numerical Study of the Effect of Gaps on Laminar Turbulent Transition of Incompressible Boundary Layers," *Procedia IUTAM*, Vol. 14, No. 0, 2015, pp. 448–458. <https://doi.org/10.1016/j.piutam.2015.03.073>, URL <http://dx.doi.org/10.1016/j.piutam.2015.03.073>.
- [10] Perraud, J., Arnal, D., and Kuehn, W., "Laminar-turbulent transition prediction in the presence of surface imperfections," *International Journal of Engineering Systems Modelling and Simulation*, Vol. 6, No. 3-4, 2014, pp. 162–170. <https://doi.org/10.1504/IJESMS.2014.063129>.

- [11] Beguet, S., Perraud, J., Forte, M., and Brazier, J. P., "Modeling of transverse gaps effects on boundary-layer transition," *Journal of Aircraft*, Vol. 54, No. 2, 2017, pp. 794–801. <https://doi.org/10.2514/1.C033647>.
- [12] Crouch, J. D., and Kosorygin, V. S., "Surface Step Effects on Boundary-Layer Transition Dominated by Tollmien–Schlichting Instability," *AIAA Journal*, Vol. 58, No. 7, 2020, pp. 2943–2950. <https://doi.org/10.2514/1.j058518>.
- [13] Crouch, J., Kosorygin, V., Sutanto, M., and Miller, G., "Characterizing surface-gap effects on boundary-layer transition dominated by Tollmien–Schlichting instability," *Flow*, Vol. 2, No. E8, 2022. URL <http://doi:10.1017/flo.2022.1>.
- [14] Zahn, J., and Rist, U., "Impact of deep gaps on laminar - Turbulent transition in compressible boundary-layer flow," *AIAA Journal*, Vol. 54, No. 1, 2016, pp. 66–76. <https://doi.org/10.2514/1.J054112>.
- [15] Mathias, M. S., and Medeiros, M. A. F., "Global instability analysis of a boundary layer flow over a small cavity," *AIAA Aviation 2019 Forum 17-21 June 2019, Dallas, Texas*, 2019.
- [16] Krishnamurty, K., "Acoustic radiation from two-dimensional rectangular cutouts in aerodynamic surfaces," Technical Note 3487, National Advisory Committee for Aeronautics, Washington, US, Aug. 1955. URL <https://digital.library.unt.edu/ark:/67531/metadc57843/>.
- [17] Plumblee, H. E., Gibson, J. S., and Lassiter, L. W., "Theoretical and Experimental Investigation of The Acoustic Response of Cavities In An Aerodynamic Flow," *Technical Report*, , No. USAF Report WADD-TR-61-75, 1962, pp. 1–167.
- [18] Rossiter, J. E., "Wind-tunnel experiments on the flow over rectangular cavities at subsonic and transonic speeds," *Reports and Memoranda No. 3438 - Aeronautical Research Council - Ministry of Aviation*, 1964.
- [19] Sarohia, V., "Experimental investigation of oscillations in flows over shallow cavities," *AIAA Journal*, Vol. 15, No. 7, 1977, pp. 984–991. <https://doi.org/10.2514/3.60739>.
- [20] Yamouni, S., Sipp, D., and Jacquin, L., "Interaction between feedback aeroacoustic and acoustic resonance mechanisms in a cavity flow: A global stability analysis," *Journal of Fluid Mechanics*, Vol. 717, 2013, pp. 134–165. <https://doi.org/10.1017/jfm.2012.563>.
- [21] Mathias, M. S., and Medeiros, M. A. F., "The effect of incoming boundary layer thickness and Mach number on linear and nonlinear Rossiter modes in open cavity flows," *Theor. Comput. Fluid Dyn.*, Vol. 35, No. 4, 2021, pp. 495–513.
- [22] Brés, G. A., and Colonius, T., "Three-dimensional instabilities in compressible flow over open cavities," *Journal of Fluid Mechanics*, Vol. 599, 2008, pp. 309–339. <https://doi.org/10.1017/S0022112007009925>.
- [23] De Vicente, J., Basley, J., Meseguer-Garrido, F., Soria, J., and Theofilis, V., "Three-dimensional instabilities over a rectangular open cavity: From linear stability analysis to experimentation," *Journal of Fluid Mechanics*, Vol. 748, 2014, pp. 189–220. <https://doi.org/10.1017/jfm.2014.126>.
- [24] Meseguer-Garrido, F., De Vicente, J., Valero, E., and Theofilis, V., "On linear instability mechanisms in incompressible open cavity flow," *Journal of Fluid Mechanics*, Vol. 752, 2014, pp. 219–236. <https://doi.org/10.1017/jfm.2014.253>.
- [25] Sun, Y., Taira, K., Cattafesta, L. N., and Ukeiley, L. S., "Biglobal instabilities of compressible open-cavity flows," *Journal of Fluid Mechanics*, Vol. 826, 2017, pp. 270–301. <https://doi.org/10.1017/jfm.2017.416>.
- [26] Mathias, M. S., and de Medeiros, M. A. F., "Interaction between rossiter and görtler modes in the compressible flow in an open cavity," *Aiaa Aviation 2020 Forum*, Vol. 1 PartF, 2020, pp. 1–6. <https://doi.org/10.2514/6.2020-3074>.
- [27] Victorino, V. B., Aguirre, F. O., and de Medeiros, M. A. F., "Gap induced boundary layer transition," *AIAA AVIATION Forum*, 2023, pp. 1–14.
- [28] Amaral, F. R., Rico, J. C. S., Bresci, C. S., Beraldo, M. M., Victorino, V. B., Gennaro, E. M., and Medeiros, M. A. F., "The Low Acoustic Noise and Turbulence Wind Tunnel of the University of Sao Paulo," *The Aeronautical Journal*, 2021, pp. 1–45.
- [29] Arnoldi, W. E., "The principle of minimized iterations in the solution of the matrix eigenvalue problem," *Quarterly of Applied Mathematics*, Vol. 9, 1951, pp. 17–29.
- [30] Lele, S. K., "Compact Finite and Difference Schemes and with Spectral-like and Resolution," *Journal of Computational physics*, Vol. 103, 1992, pp. 16–42.
- [31] Gaitonde, D., and Shang, J. S., "High-order finite-volume schemes in wave propagation phenomena," *27th Plasma Dynamics and Lasers Conference*, , No. June, 1996. <https://doi.org/10.2514/6.1996-2335>.

- [32] Gaitonde, D. V., and Visbal, M. R., “High-Order Schemes for Navier-Stokes Equations: Algorithm and Implementation Into FDL3DI,” Technical Report 1998-3060, AIR FORCE RESEARCH LAB WRIGHT-PATTERSON, Jul. 1998. URL <https://apps.dtic.mil/sti/citations/ADA364301>.
- [33] Åkervik, E., Brandt, L., Henningson, D. S., Høpfner, J., Marxen, O., and Schlatter, P., “Steady solutions of the Navier-Stokes equations by selective frequency damping,” *Physics of Fluids*, Vol. 18, No. 6, 2006, p. 068102. <https://doi.org/10.1063/1.2211705>.
- [34] Li, N., and Laizet, S., “2DECOMP&FFT - A Highly Scalable 2D Decomposition Library and FFT Interface,” Tech. rep., Numerical Algorithms Group (NAG) - Imperial College London, 2010.
- [35] Tezuka, A., and Suzuki, K., “Three-dimensional global linear stability analysis of flow around a spheroid,” *AIAA Journal*, Vol. 44, No. 8, 2006, pp. 1697–1708. <https://doi.org/10.2514/1.16632>.
- [36] Mathias, M. S., “Computational study of the hydrodynamic stability of gaps and cavities in a subsonic compressible boundary layer,” Ph.D. thesis, São Carlos School of Engineering, University of São Paulo, 2021.
- [37] Crouch, J., Kosorygin, V., Sutanto, M., and Miller, G., “Characterizing surface-gap effects on boundary-layer transition dominated by Tollmien–Schlichting instability,” *Flow*, Vol. 2 (E8), 2022. <https://doi.org/10.1017/flo.2022.1>.
- [38] Veerasamy, D., and Atkin, C., “A rational method for determining intermittency in the transitional boundary layer,” *Experiments in Fluids*, Vol. 61, No. 1, 2020, pp. 1–13. <https://doi.org/10.1007/s00348-019-2856-5>.